Every term of OEIS A395833 after the second is divisible by 3

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Abstract

OEIS A395833 is defined by the vanishing condition $[x^n]A\big(x/A(x)^{2n-1}\big) = 0$ for $n > 1$, and carries a conjecture of P. D. Hanna that $a(n) \equiv 0 \pmod 3$ for $n \geq 2$. It is true, and the proof reduces the whole statement to a single divisibility of binomial coefficients. The condition determines the coefficients one at a time, with $a(n)$ entering with coefficient 1, so it determines them modulo 3; and modulo 3 the *polynomial* $1 + x$ satisfies the condition, because substituting it turns the n-th condition into $(-1)^{n-1}\binom{3n-3}{n-1} \equiv 0 \pmod 3$, which Legendre's formula gives outright: $v_3\big(\binom{3m}{m}\big) = \frac{1}{2}s_3(2m) \geq 1$ for every $m \geq 1$. Uniqueness then forces $A(x) \equiv 1+x$, which is the conjecture. The entry's other conjecture, that $2n-1$ divides $a(n)$, is *not* settled here.

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1 The sequence and the conjecture

OEIS A395833 is "G.f. A(x) satisfies [x^n] A(x/A(x)^(2*n-1)) = 0 for n > 1.". It has offset 0 and begins

$$1, 1, 3, 30, 567, 16452, 663894, 35188218, \ \ldots$$

The entry gives the expansion

G.f.: A(x) = 1 + x + 3*x^2 + 30*x^3 + 567*x^4 + 16452*x^5 + 663894*x^6 + ...

so, writing $A(x) = \sum_{n\geq 0} a(n)x^n$, the entry fixes

$$a(0) = 1, \qquad a(1) = 1, \tag{1}$$

and defines the remaining coefficients by

$$[x^n]\, A\left(\frac{x}{A(x)^{2n-1}}\right) \;=\; 0 \qquad \text{for } n > 1. \tag{2}$$

Since $a(0) = 1$, the series A is invertible and $A^{-(2n-1)}$ is a well-defined element of $\mathbb{Z}[[x]]$, so the composition in (2) makes sense: the inner series $x\,A(x)^{-(2n-1)}$ has zero constant term. This is the only input taken from the entry.

The entry separately carries the following comment:

Conjecture: a(n) == 0 (mod 3) for n >= 2. — *Paul D. Hanna*

As of the "Last modified" line on the live entry (May 09 2026 00:29:02, revision 7) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way. The entry carries a second conjecture of the same contributor, that $2n-1$ divides $a(n)$ for $n \geq 1$. It is not proved below and it is not claimed.

2 The condition determines the sequence

Lemma 1. *Given* (1)*, there is exactly one sequence of integers* $a(2), a(3), \ldots$ *satisfying* (2)*, and for each* $N \geq 2$ *the condition at* $n = N$ *has the form* $a(N) =$ (*a polynomial with integer coefficients in* $a(0), \ldots, a($
$1)$)*.*

Proof. Fix $N \geq 2$ and write $u = x\,A(x)^{-(2N-1)}$, so $u = x + O(x^2)$ and $[x^m]u^j = 0$ for $m < j$, while $[x^j]u^j = 1$. Expanding,

$$[x^N]A(u) \;=\; \sum_{j\geq 0} a(j)\,[x^N]u^j \;=\; \sum_{j=0}^{N} a(j)\,[x^N]u^j.$$

The term $j = N$ contributes $a(N)\cdot 1$. For $j < N$ the factor $[x^N]u^j = [x^{N-j}]A^{-j(2N-1)}$ is a polynomial in $a(1), \ldots, a(N-j)$ with $N-j \leq N-1$, and it is multiplied by $a(j)$ with $j \leq N-1$; so no $a(N)$ occurs there. Hence the coefficient of $a(N)$ in (2) is exactly 1, the condition solves for $a(N)$ over $\mathbb{Z}$, and induction from (1) gives existence and uniqueness. □

Corollary 2. *Let* $k \geq 2$*. If* $\bar{A} \in (\mathbb{Z}/k\mathbb{Z})[[x]]$ *has* $[x^0]\bar{A} = [x^1]\bar{A} = 1$ *and satisfies* (2) *with all coefficients read modulo* k*, then* $a(n) \equiv [x^n]\bar{A} \pmod{k}$ *for every* n*.*

Proof. Reduction modulo k is a ring homomorphism and carries the recursion of Lemma 1 to the same recursion over $\mathbb{Z}/k\mathbb{Z}$, where the coefficient of $a(N)$ is still 1 and hence invertible. (The inverse $A^{-(2N-1)}$ is carried to the inverse of the reduction, which exists because the constant term 1 is a unit.) That recursion, started from the same two initial values, has a unique solution, and both $a \bmod k$ and the coefficients of $\bar{A}$ solve it. □

3 A divisibility of binomial coefficients

For a prime p let $s_p(m)$ denote the sum of the base-p digits of m, and v_p the p-adic valuation.

Lemma 3. $v_3\left(\binom{3m}{m}\right) = \dfrac{s_3(2m)}{2}$ *for every* $m \geq 0$*. In particular* 3 *divides* $\binom{3m}{m}$ *for every* $m \geq 1$*.*

Proof. Legendre's formula gives $v_3(j!) = \big(j - s_3(j)\big)/2$. Hence

$$v_3\left(\binom{3m}{m}\right) = v_3\big((3m)!\big) - v_3\big(m!\big) - v_3\big((2m)!\big) = \frac{\big(3m - s_3(3m)\big) - \big(m - s_3(m)\big) - \big(2m - s_3(2m)\big)}{2},$$

which simplifies to $\big(s_3(m) + s_3(2m) - s_3(3m)\big)/2$. Multiplying by 3 shifts the base-3 digits and appends a zero, so $s_3(3m) = s_3(m)$ and the expression collapses to $s_3(2m)/2$.

For $m \geq 1$ we have $2m \geq 2$, so $s_3(2m) \geq 1$; since the valuation is an integer, $s_3(2m)/2 \geq 1$, that is $3 \mid \binom{3m}{m}$. □

4 The reduction modulo 3

Lemma 4. *Over* $\mathbb{Z}/3\mathbb{Z}$ *the polynomial* $\bar{A}(x) = 1 + x$ *satisfies* (1) *and* (2)*.*

Proof. Clearly $[x^0]\bar{A} = [x^1]\bar{A} = 1$. Fix $n > 1$ and put $u = x\,(1+x)^{-(2n-1)}$. Since $\bar{A}(u) = 1 + u$,

$$[x^n]\bar{A}(u) \;=\; [x^n]u \;=\; [x^{n-1}](1+x)^{-(2n-1)}.$$

By the binomial series, $[x^m](1+x)^{-k} = (-1)^m\binom{k+m-1}{m}$, so with $k = 2n-1$ and $m = n-1$,

$$[x^n]\bar{A}(u) \;=\; (-1)^{n-1}\binom{(2n-1)+(n-1)-1}{n-1} \;=\; (-1)^{n-1}\binom{3n-3}{n-1} \;=\; (-1)^{n-1}\binom{3m'}{m'}, \quad m' = n-1.$$

Since $n > 1$ we have $m' \geq 1$, so Lemma 3 makes this $\equiv 0 \pmod 3$. Hence (2) holds modulo 3. $\square$

Theorem 5. $a(n) \equiv 0 \pmod 3$ *for every* $n \geq 2$. *This is the conjecture of Section 1.*

Proof. By Lemma 4 the polynomial $1 + x$ satisfies (1) and (2) over $\mathbb{Z}/3\mathbb{Z}$, so by Corollary 2 it is the reduction of A modulo 3. Its coefficients are $1, 1, 0, 0, \ldots$, so $a(n) \equiv 0 \pmod 3$ for every $n \geq 2$. $\square$

Remark 6. The proof is short because the reduction is a polynomial rather than an infinite series: the whole content is that the substitution $x \mapsto x(1+x)^{-(2n-1)}$ makes the n-th coefficient a single central-type binomial, and that binomial is always divisible by 3. The same computation modulo 2 gives $\binom{3n-3}{n-1}$ modulo 2, which is *not* always even — for instance $\binom{6}{2} = 15$ — so $1 + x$ is not the reduction modulo 2 and the argument is specific to the prime 3.

5 Verification

Three checks, each independent of the algebra above.

First, the recursion of Lemma 1 was run in exact rational arithmetic from (1), inverting A by the usual convolution, and its output compared against the entry's DATA at every published index within reach; the values agree exactly and are integers. At every step the coefficient of $a(N)$ was computed rather than assumed, by evaluating the condition at $a(N) = 0$ and at $a(N) = 1$ and subtracting; it came out 1 every time.

Second, Lemma 3 was checked directly: $3 \mid \binom{3m}{m}$ for every $m \leq 400$, and $v_3\left(\binom{3m}{m}\right) = s_3(2m)/2$ over the same range, together with the identity $s_3(3m) = s_3(m)$ used in its proof.

Third, the reduced condition of Lemma 4 was evaluated for every n from 2 to 119: the value $(-1)^{n-1}\binom{3n-3}{n-1}$ is divisible by 3 in every case, with no exception.

References

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